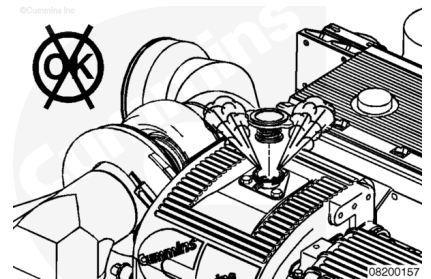


Trainings ▾

Drain

⚠ WARNING ⚠

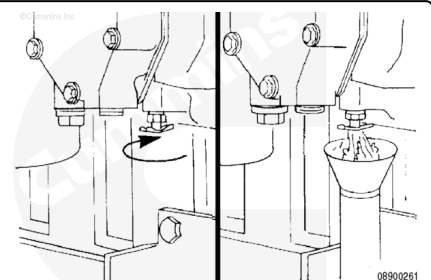
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the C Series engine cooling system by opening the drain valve on the engine oil cooler (exhaust side of engine). A drain pan with a capacity of 31.7 liters (8 gal) will be adequate in most installations. Remove the pressure cap to allow the coolant to drain properly. After the cooling system is completely drained, close the drain valve.



Flush

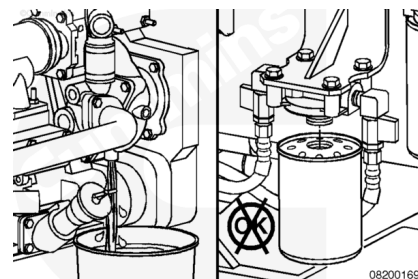
RESTORE™ is a heavy-duty cooling system cleaner that removes corrosion products, silica gel, and other deposits. The performance of RESTORE™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times or the use of RESTORE Plus™. Up to twice the recommended concentration levels of RESTORE™ can be used



safely. RESTORE Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

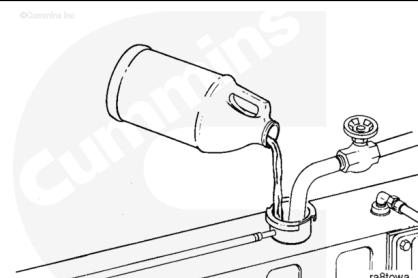


If not previously done, drain the cooling system. Refer to Cooling System - Drain. Do **not** allow the cooling system to dry out.

Do **not** remove the coolant filter.

⚠ CAUTION ⚠

Fleetguard® RESTORE™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operations.



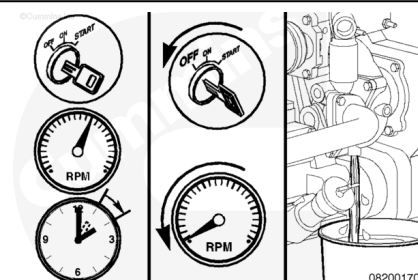
⚠ CAUTION ⚠

The system must be filled properly to prevent air locks. During filling, air must be purged from the engine coolant passages. Be sure to open the petcock on the aftercooler for aftercooled engines. Wait 2 to 3 minutes to allow air to be vented; then add mixture to bring the level to the top.

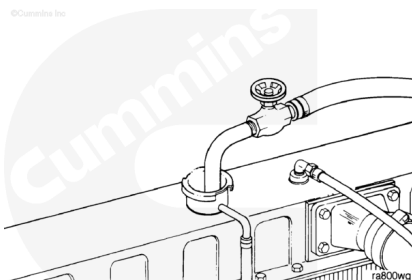
Immediately add 3.8 liters [1 gal] of Fleetguard® RESTORE™, RESTORE Plus™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity. Fill the system with clean water.

Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1½ hours.

Shut off the engine, allow to cool to 50°C [122°F] and drain the cooling system.



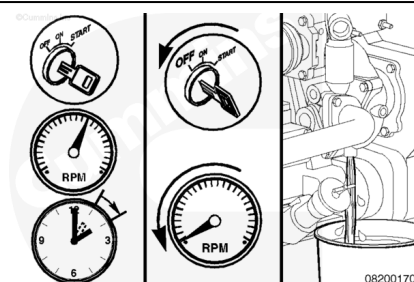
Fill the cooling system with clean water.



Operate the engine for 5 minutes with the coolant temperature above 85°C [185°F].

Shut off the engine, allow to cool to 50°C [122°F], and drain the cooling system.

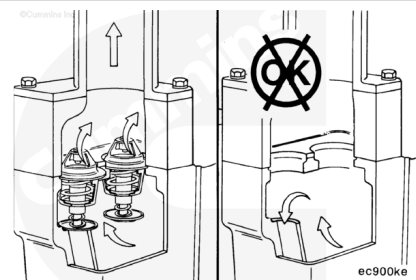
If the water being added is still dirty, the system **must** be flushed again until the water is clean.



Fill

⚠ CAUTION ⚠

The system must be filled properly to prevent air locks. During filling, air must be purged from the engine coolant passages. Be sure to open the petcock on the aftercooler for aftercooled engines. Wait 2 to 3 minutes to allow air to be vented; then add water to bring the level to the top.



Fill the cooling system with fully formulated coolant or a 50/50 mixture of recommended antifreeze, and clean, good-quality water. Refer to Procedure 018-004

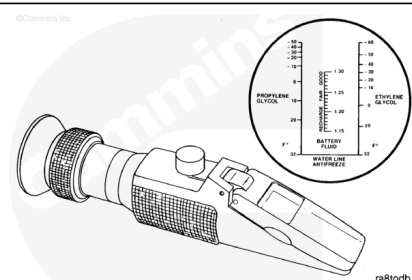
([/qs3/pubsys2/xml/en/procedures/41/41-018-004.html](http://qs3/pubsys2/xml/en/procedures/41/41-018-004.html))

(Coolant Recommendation and Specifications) in Section V.

Install the pressure cap. Operate the engine to 50°C [122°F], and check for coolant leaks.

The freeze point protection **must** be checked after coolant is added to the cooling system.

The Fleetguard® refractometer, Part Number C2800, provides a reliable, easy-to-read, and accurate measurement of freeze point protection and antifreeze concentration.



Refer to the OEM's instructions for the correct operation instructions for the Fleetguard® refractometer, Part Number C2800.

Last Modified: 15-Jul-2005
